

Part III · AI Exercise 3

Literature Synthesis and Historical Context

3(a)

These are the values reported in **Part I(b), Table 1**. Estimates and standard errors are in **percent per month**; the table's t-statistics test against zero.

Sample	Parameter	Estimate	Standard error	t	p
July 1969–June 2026	γ_0	0.6119	0.2191	2.79	0.005
July 1969–June 2026	γ_M	0.0352	0.2725	0.13	0.897
July 1926–June 2026	γ_0	0.5160	0.1714	3.01	0.003
July 1926–June 2026	γ_M	0.2490	0.2315	1.08	0.282

The reported mean market excess returns are **0.6116%** and **0.6952%** per month, respectively. For the main sample, your report gives **t = -2.12** for testing the slope against the sample market premium.

My Part I(b) results weaken the favorable evidence of Fama and MacBeth (1973) and contradict the Sharpe-Lintner CAPM restrictions in my sample. For July 1969–June 2026, $\hat{\gamma}_0 = 0.6119\%$ per month (SE = 0.2191, t = 2.79, p = 0.005), significantly above the zero intercept predicted for excess returns. Meanwhile, $\hat{\gamma}_M = 0.0352\%$ per month (SE = 0.2725, t = 0.13, p = 0.897), providing no statistically significant evidence of a positive beta premium. The slope is also far below the CAPM benchmark of 0.6116%, with the reported benchmark test rejecting equality at the 5% level (t = -2.12). Thus, the estimated security market line is nearly flat, failing to replicate their positive beta-return relationship. The longer July 1926–June 2026 sample reinforces this pattern: the intercept remains significantly positive, while the beta premium remains insignificant. A positive excess-return intercept alone does not reject Black's zero-beta CAPM, but the weak beta-return relationship undermines the earlier favorable interpretation.

3(b)

The original LLM response correctly identifies Banz's (1981) size effect: small firms earned higher average returns than their market betas could explain. It also accurately summarizes Fama and French (1992), emphasizing the explanatory power of size and book-to-market and the weak relation between beta and average returns once beta variation is separated from size. However, it omits Rosenberg, Reid, and Lanstein (1985), an important earlier study documenting the book-to-market effect that the assignment explicitly asks about. Although its discussion of Fama and French (1993) adds useful context, it does not fill this omission. Overall, the response captures the main empirical challenges to the CAPM but gives an incomplete account of the development of the value evidence.

3(c)

The original LLM response does not acknowledge data snooping: size and book-to-market may appear important partly because researchers identified and tested these patterns using the same historical data. This concern also applies to Part I(f), since our sample overlaps with the data used to discover these effects, so it is not a fully independent test. However, we use previously documented characteristics rather than search across many predictors ourselves. Our results provide only weak evidence: size is insignificant ($t = 0.86$), while book-to-market is marginally significant ($t = 1.95$). Lagging these characteristics addresses their timing but does not eliminate data snooping. Testing on a separate period after the original studies would provide stronger independent evidence.

3(d)

LLM response to record

International evidence shows that size and value effects occur outside the United States, although their strength varies across markets and samples.

- **Direct Fama-MacBeth evidence:** Hu, Chen, Shao, and Wang (2019) find a significant size-factor premium in Chinese stocks, while market and value-factor premia are insignificant. Thus, international Fama-MacBeth tests do not uniformly support both size and value. ^[1]

- **Broader international evidence:** Rouwenhorst (1999) finds that small stocks outperform large stocks and value stocks outperform growth stocks in emerging markets. Fama and French (1998) find value outperformance in twelve of thirteen major markets. These studies provide complementary evidence from portfolio comparisons and factor tests; their findings should not all be described as Fama-MacBeth regression results. ^{[2], [3]}

My interpretation is that international replication reduces the likelihood that these effects are merely artifacts of searching U.S. data. It is consistent with compensation for omitted risks, but does not establish that explanation: similar investor biases and limits to arbitrage could also produce recurring patterns across countries.

Evaluation for submission

The international evidence modestly strengthens the plausibility of a risk-based interpretation by showing that size and value patterns are not confined to U.S. data. However, it strengthens the evidence that these patterns exist more clearly than it distinguishes risk from mispricing, since investor biases and limits to arbitrage may also operate internationally. The LLM appropriately acknowledges this distinction and the variation across markets. In our Part I(f), size is insignificant ($t = 0.86$) and book-to-market is only marginally significant ($t = 1.95$), so international findings cannot turn our weak evidence into proof of priced risks. They provide supporting context for the positive book-to-market coefficient, but establishing a risk explanation would require evidence linking returns to exposure to economically meaningful risks.

Sources cited in 3(d)

[1] Hu, Chen, Shao, and Wang (2019). "Fama-French in China: Size and Value Factors in Chinese Stock Returns." doi:10.1111/irfi.12177

[2] Rouwenhorst (1999). "Local Return Factors and Turnover in Emerging Stock Markets." doi:10.1111/0022-1082.00151

[3] Fama and French (1998). "Value versus Growth: The International Evidence." doi:10.1111/0022-1082.00080